Monetary Tightening Shocks and Opioid-Related Harms across Canada: A Province by Quarter Analysis

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Abstract

Canada continues to combat the fatal use of opioids across all provinces despite efforts to unravel underlying reasons that may contribute to the crisis. This study examines whether macroeconomic factors, specifically Bank Of Canada interest rate hikes, are associated with subsequent changes in opioid-related deaths and hospitalizations across the nation. Using a province-by-quarter panel from 2016Q1 to 2025Q2, we combined provincial opioid-harm data with the Bank of Canada policy rate and quarterly provincial unemployment. For our study, we defined a "tightening-shock" as an increase in 0.50 percentage points relative to the previous quarter; and examined whether there was change in opioid harms one to two quarters later. Our regression results indicated that interest rate hikes under our threshold were jointly associated with increases in opioid-related deaths when analyzed under our unemployment control model, while results for hospitalizations were not statistically significant under any model. These findings could be helpful for public health awareness in metro-governments and could potentially be a factor to anticipate and mitigate opioid harms during periods of aggressive policy hikes.

Introduction

Despite the country's developed healthcare and social support systems, opioid use remains highly prevalent in Canada. (Belzak & Halverson, 2018). In many ways, opioid use appears difficult to eliminate entirely, as supply and demand persist even under strict policy conditions. Locally, British Columbia has stood out due to its recent approach to drug policy, including the decriminalization of certain substances such as fentanyl, morphine, and methamphetamine. Policies such as these highlight the ongoing effort, both within Canada and globally, to reduce opioid use and its associated harms (Belzak & Halverson, 2018). People use opioids for a variety of reasons, including curiosity, peer influence, and stress relief. As a result, many strategies aim to reduce the perceived appeal of opioid use, particularly among younger populations. At the same time, efforts have been made to reduce stigma, with the goal of allowing individuals to access support services without fear of judgment (Public Health Agency of Canada, 2023).

Despite these efforts, opioid harm-reduction and rehabilitation programs do not fully meet existing demand. In practice, individuals seeking treatment may face long wait times, and emergency responses are not always able to prevent fatal outcomes (Public Health Agency of Canada, 2023). This suggests that while current policies and interventions are important, they may not be sufficient on their own. Rather than focusing only on reducing supply or improving treatment access, it may also be useful to consider how resources are allocated and whether broader factors influence when and where harms occur. In analyzing the data, a clear pattern emerges: periods of increasing economic pressure appear to align with increases in opioid-related harms.

Opioid-related harms continue to be a major public health issue in Canada, particularly in

the form of overdose deaths and hospitalizations (Belzak & Halverson, 2018; Statistics Canada, 2026). Over the past decade, both measures have shown concerning trends, suggesting that the problem is not only persistent but also evolving over time. Most explanations for these harms focus on direct factors such as drug supply and policy, as well as prescribing practices. While these are clearly important, they do not fully explain why harms rise and fall over time. This suggests that broader, system-level factors may also be influencing outcomes.

One possible factor is the overall economic environment and pressure. When economic conditions worsen, individuals may experience increased financial strain, job insecurity, and reduced access to resources. These conditions can lead to higher stress levels, which in turn might increase the likelihood of substance use as a coping mechanism (Braveman & Gottlieb, 2014). In addition, barriers to healthcare can make it more difficult for individuals to seek treatment or support. Taken together, these effects suggest that economic stress contributes to worsening health outcomes, including opioid-related harms.

One way to observe changes in economic conditions is through monetary policy. In the case of this study, the policies in question are interest rate changes set by institutions like the Bank of Canada (Bank of Canada, n.d.). Unlike broader economic indicators like unemployment or income, interest rate changes occur at clearly defined points in time and are directly set by policy. This makes them easier to identify as discrete shocks, allowing for a more precise analysis of how sudden increases in financial pressure may translate into changes in opioid-related harms.

When interest rates increase and borrowing becomes more expensive, both households and businesses may face tighter financial conditions. Following the same logic as before, the resulting financial stress can cause mental strain, which in turn leads to a higher likelihood of

substance use, and therefore an increased risk of opioid-related harm (Case & Deaton, 2020).

However, these effects are not immediate. Just as it takes time for interest rate changes to affect spending and employment, it likely also takes time for these changes to influence health outcomes. A rate increase today may not significantly impact an individual's financial situation until months later, once expenses accumulate or income becomes unstable. Because of this, any relationship between interest rates and opioid harms is expected to appear with a delay, which this study estimates to be on the order of one to two quarters.

Despite this plausible mechanism, there is little existing research examining the relationship between interest rate changes and opioid-related harms directly, especially in Canada. Most studies focus on long-term socioeconomic differences and economic cycles rather than short-term policy changes. This creates a gap in the current literature: if rapid increases in interest rates act as identifiable shocks, they may provide a new way to test whether economic stress translates into measurable changes in health outcomes.

This research paper tests whether rapid increases in interest rates are followed by increases in opioid-related harms. The expectation is that any effect will not be immediate, but will instead appear after one to two quarters.

Methods

Our study required the use of three datasets. Firstly, we obtained our "Substance Harms" data which we filtered to only include opioid observations reported per quarter for the ten Canadian provinces under our specified timeline. (Public Health Agency of Canada, 2026). Next we added the national policy rate data which was obtained from the Bank of Canada Valet series and aggregated the values to be quarterly. (Bank of Canada, n.d.). Lastly, we have our unemployment dataset which we converted to quarterly averages to be able to fit in the same

time window. (Statistics Canada, 2026). This dataset was included in our study as a control model because unemployment is a major alternative explanation for changes in opioid harms that has the potential to move at the same time as interest rates. Overall, all datasets were cleaned using R or Python and standardized to a consistent time identifier (YYYYQ#); then merged into a master province-quarter panel.

The primary exposure variable was a "monetary tightening shock", which we defined as a quarter in which the interest rate increased by at least 0.50 percentage points from the previous quarter. This threshold was set because a standard increase quarter-to-quarter is typically 25 basis points or 0.25%. Our threshold for a "shock" is essentially two times the anticipated increase. Using this information, we test delayed associations by defining two new variables called *lag1* and *lag2* which respectively represent the change in opioid-harms 1-2 quarters subsequent to an interest rate hike. With these variables, we estimated multiple regression models separately for both opioid deaths and opioid-related hospitalizations. Our models include province fixed effects to control for invariance and a time trend to account for overall changes across our study period. This ensures that we are comparing provinces fairly and controls stable differences such as population size or long-run differences in health systems. Since interest rate shocks are a national measure that varies only across time, full quarter fixed effects could not be included without absorbing the shock variables. Following the regression, we conducted a joint F-test of the lagged shock coefficients at $\alpha = 0.05$. We indicated a null hypothesis (H_0:$\beta_{lag1} = 0$ and $\beta_{lag2} = 0$) that set both lag effects equal to zero while our alternative hypothesis (H_A:$\beta_{lag1} \neq 0$ or $\beta_{lag2} \neq 0$) was that at least one of the effects differed from zero. Overall, this design allows us to evaluate lagged associations between tightening shocks and opioid outcomes while controlling for differences between the provinces.

Results

We begin with descriptive visualizations of opioid outcomes and monetary tightening over time, and then present regression-based hypothesis tests of lagged shocked effects. Here, Figure 1 shows quarterly opioid deaths and hospitalizations across Canadian provinces from 2016Q1 to 2025Q2.

Figure 1: Opioid Harms over Time

Both outcomes increased over the observation period, with deaths rising from roughly 1000 per quarter in 2016 to over 2000 by the early 2020s before showing some fluctuation in later years. Hospitalizations followed a similar upward pattern, though with a somewhat flatter trajectory compared to deaths. The consistent upward trend across both outcomes motivated the inclusion of a linear time trend in all regression models, to separate pre-existing growth from any effects attributable to monetary tightening shocks.

Figure 2 plots the Bank of Canada overnight rate alongside the timing of identified tightening shocks; quarters where the rate increased by 50 basis points or more relative to the prior quarter.

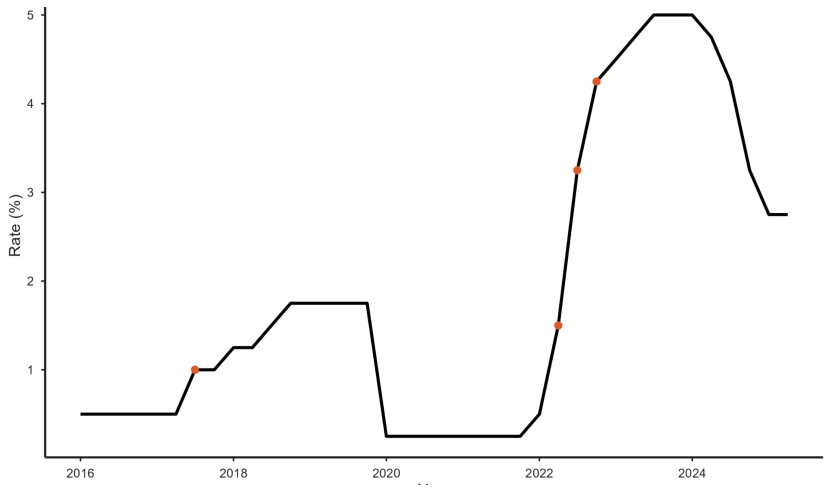

Figure 2: Interest Rate/Tightening Shocks Over Time (Dots represent shocks)

The policy rate remained low and largely stable from 2016 through 2021, before rising sharply through 2022 and into 2023. The majority of identified shocks cluster within this tightening cycle, with a smaller number of shocks appearing in earlier periods. This pattern confirms that the shock variable captures distinct, concentrated episodes of rapid monetary tightening rather than gradual or routine rate adjustments, and provides meaningful variation for testing lagged associations with opioid outcomes.

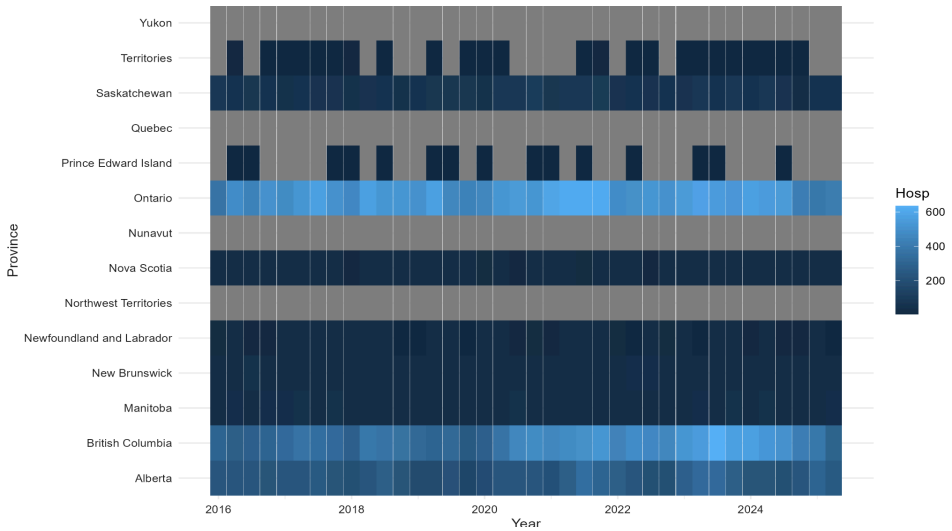

Figure 3: Provincial Variation in Opioid Harms

Lighter shading indicates higher hospitalization counts, with values ranging from approximately 200 in darker cells to 600 in the lightest. Substantial variation is visible both across provinces and within provinces over time. Some provinces show consistently lighter shading throughout the study period, while others remain darker. This cross-provincial variation supports the use of province fixed effects in the regression models, as it indicates that each province carries a distinct baseline level of opioid-related harm that needs to be accounted for independently of the shock exposure.

Table 1 reports coefficient estimates and joint F-test results from four panel regression models, with opioid deaths and hospitalizations as separate outcomes, each estimated with and without unemployment as a control variable.

Table 1: Regression Results and Joint F-Test (α=0.05)

Outcome	Model	Shock (t-1) β (p)	Shock (t-2) β (p)	Unemp β (p)	Joint F-test p	Decision (α=0.05)
Deaths	Baseline	7.56 (0.602)	23.12 (0.115)	—	0.134	Retain H0
Deaths	+ Unemployment	18.78 (0.225)	27.08 (0.075)	**11.58 (<0.001)**	**0.0246**	Reject H0
Hospitalizations	Baseline	-1.04 (0.904)	4.84 (0.572)	—	0.846	Retain H0
Hospitalizations	+ Unemployment	1.41 (0.886)	6.90 (0.475)	2.84 (0.122)	0.700	Retain H0

All of our models include province fixed effects (Table 1). Regarding opioid deaths, our baseline model without unemployment produced positive coefficients for both shock variables, indicating an increase in deaths/hospitalizations; though neither reached statistical significance at $\alpha = 0.05$, and failing to reject the null hypothesis. After adding unemployment as a control, the lag estimates increased and the two-quarter lag was marginal, while unemployment was strongly significant. Importantly, the joint F-test rejected the null hypothesis for the unemployment-deaths model ($p = 0.0246$), indicating that the lagged shock terms are jointly associated with deaths in that specification. In contrast, hospitalization models showed little evidence of an association:

lag coefficients were small and non-significant, unemployment was not significant, and joint F-tests retained the null hypothesis in both models (Table 1).

Across all four models, the two-quarter lag coefficient was consistently larger than the one-quarter lag, suggesting that any association between monetary tightening and opioid harms builds over time rather than appearing immediately following a shock.

Discussion

Our results suggest that monetary tightening shocks are associated with subsequent increases in opioid harms, with the clearest evidence appearing for opioid deaths and with effects most apparent around the two quarters after. In contrast, we do not find statistically significant evidence of lagged tightening effects on opioid-related hospitalizations under the 0.50% threshold definition. We intend that our findings will be used by health care workers to further prepare for opioid incidents at more appropriate periods of time. In doing so, our healthcare safety net can be effectively deployed at the right times. Plenty of effort and funding has gone into our first responding health care system at all three levels of government; the resources aren't what we’re criticizing. Rather, using those same resources, we’ve curated a strategy that can mitigate what we view to be the inevitable harms of opioids. Interest rate hikes are also influenced by forces out of the realm of this study, so our plan is an effort to balance these two unavoidable circumstances to create a safety plan.

In stating this, it’s imperative to draw a clear distinction between correlation and causation in our report. Given that there is no one-to-one correlation between opioid harms and interest rate shocks, we cannot claim causation. Our correlation is significant and can be used to pre-emptively prepare our health care teams.

The dataset we used, “Substance Harms Dataset”, fails to address all health incidents. There are limitations that skew our data towards the underrepresentation of individuals. This dataset, from 2016-2025, is managed by the *Public Health Agency of Canada (PHAC)*. Deaths are from the *Canadian Coroner and Medical Examiner Base.*

Reporting of these two variables deaths and hospitalizations vary in ambiguity. To start, many opioid users are often marginalized/underrepresented and consequently don’t even make it into the consideration of death by opioid metrics.

Deaths are more direct, and are reported immediately with urgency, especially for publicly scrutinized reports like the Hospital Standardized Mortality Ratio.. For these reasons, deaths are reported by the National Substance Harms Dataset with higher priority. The signifiers of the health crisis put emphasis on more severe incidents. e.g. someone with a slow heart rate will not be reported in statistics as fast as a death.

Deaths that were caused by opioids may be entered as having an “unknown cause”, when the opioid was mixed with other narcotics, or the influence of opioids towards death was not put into enough consideration.

Reports of hospitalization are also skewed. Some users may not recognize the need to be hospitalized and continue to live with health risks. Another case may be users who recognize the need to be hospitalized and simply choose not to, as it impedes their freedom and opioid consumption. Others users may have people they are obliged to take care of, and therefore can’t afford to go to the hospital and go through withdrawal programs.

These limitations may explain the stronger signal for deaths than hospitalizations as underreporting and misclassification may bias results toward weaker effects. Future studies should incorporate other indicators to better capture non-fatal opioid outcomes.

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